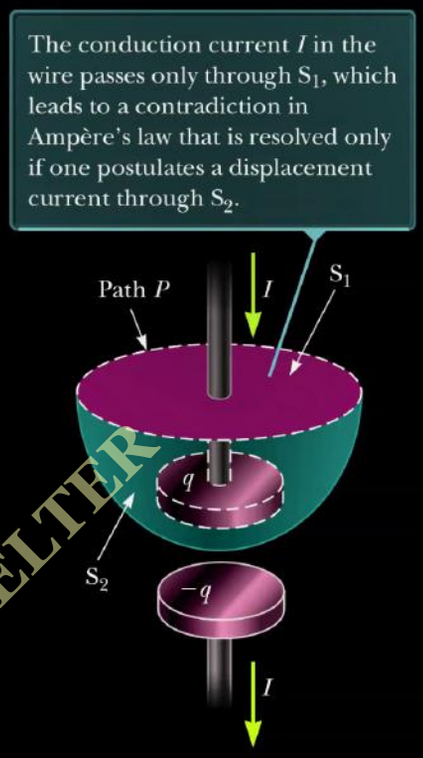
Displacement Current And Generalized Ampere's Law:

Recall the during our study of magnetics we had mostly used Ampere's law equation:

$$\int \vec{B} \cdot d\vec{l} = \mu_0 i_{enc} \quad (1)$$

Here i_{enc} was the current crossing the Amperian surface (A surface mounted on Amperian loop). It turns out that this law is incomplete and hence lead to contradictions in some situations. For instance consider a current I flowing in to a capacitor. On Amperian path P we can consider two different mountings of Amperean surfaces. For the lollipop colored surface (circular disc), a current is crossing while for the green surface (Bowl shaped), there is no current crossing. However since both surfaces are mounted on same Amperian loop, $\oint \vec{B} \cdot d\vec{l}$ would be same for both of them. This means that there is something missing from the Ampere's law in the form we mentioned. Maxwell solved this problem by introducing another term on the RHS of equation (1) which includes a factor called the displacement current and is given by:



$$i_d = \epsilon_0 \frac{d\Phi_E}{dt}$$

Here Φ_E represents the electric flux through the Amperian surface, clearly from the green surface flux is changing with time while there is no conduction current crossing it. We can show that displacement current associated with green surface is exactly equal to the conduction current through the lollipop colored surface. Recall that field between capacitor plates is $q/A\epsilon_0$. Thus if A is the area of the plates, flux associated with

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green surface is simply : $\phi_E = \left(\frac{q}{A\epsilon_0}\right) \times A = q/\epsilon_0$

$\Rightarrow \epsilon_0 (d\phi_E/dt) = dq/dt = i$ which is the conduction current passing through the disc like surface. Now if we write Ampere's law modified by Maxwell as:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 (i + i_d)$$

then the contradiction is resolved. Thus we have a fundamental law of electrodynamics in the form of Ampere - Maxwell law i.e:

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left[i + \epsilon_0 \frac{d\phi_E}{dt} \right]$$

MAXWELL'S EQUATIONS: Having described Ampere Maxwell law above, now we are in a position to write equations governing electrodynamic fields in space. Thus we have:

Gauss's law ▶	$\oint \vec{E} \cdot d\vec{A} = \frac{q}{\epsilon_0}$	$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\varepsilon_0}$
Gauss's law in magnetism ▶	$\oint \vec{B} \cdot d\vec{A} = 0$	$\vec{\nabla} \cdot \vec{B} = 0$
Faraday's law ▶	$\oint \vec{E} \cdot d\vec{s} = -\frac{d\Phi_B}{dt}$	$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$
Ampère–Maxwell law ▶	$\oint \vec{B} \cdot d\vec{s} = \mu_0 I + \epsilon_0 \mu_0 \frac{d\Phi_E}{dt}$	$\vec{\nabla} \times \vec{B} = \mu_0 \vec{j} + \frac{1}{c^2} \frac{\partial \vec{E}}{\partial t}$

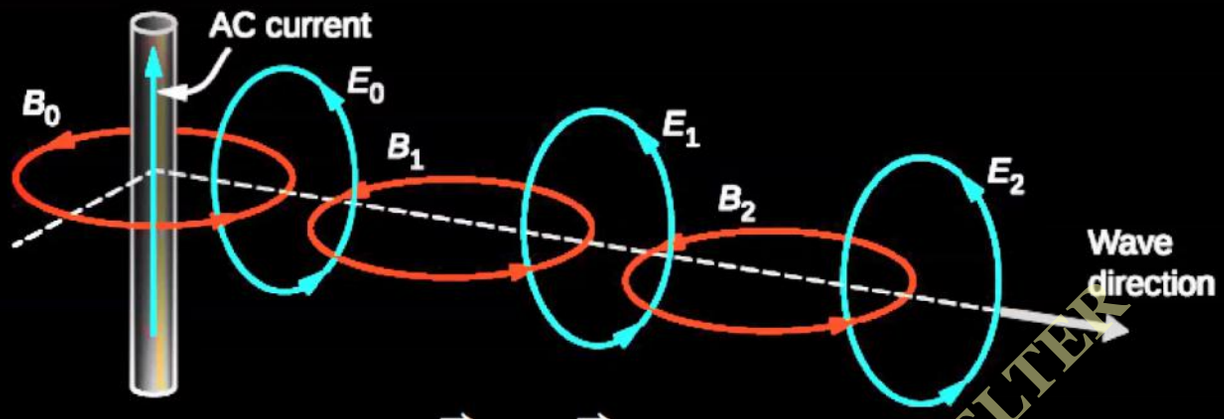
The table gives the integral and differential forms of the four Maxwell's Equations representing fundamental laws of electrodynamic in space. (In magnetic materials and dielectrics, these laws have to be further modified).

→ The Faraday's law describes creation of Electric field due to changing magnetic flux while Ampere-Maxwell law describes creation of magnetic field due to changing electric flux.

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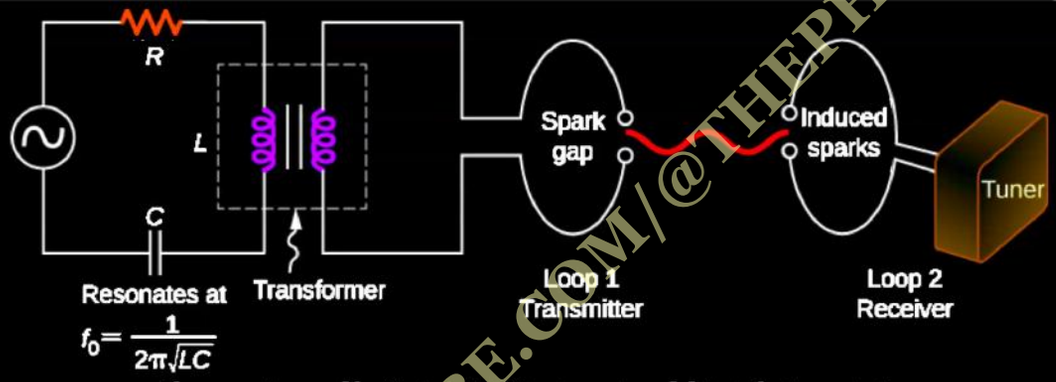
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→ From these equations we can infer that a time varying magnetic field produces an electric field which in turn can produce time varying magnetic field if varied suitably. This forms the basis of Generation of Electromagnetic waves as illustrated in the figure below:

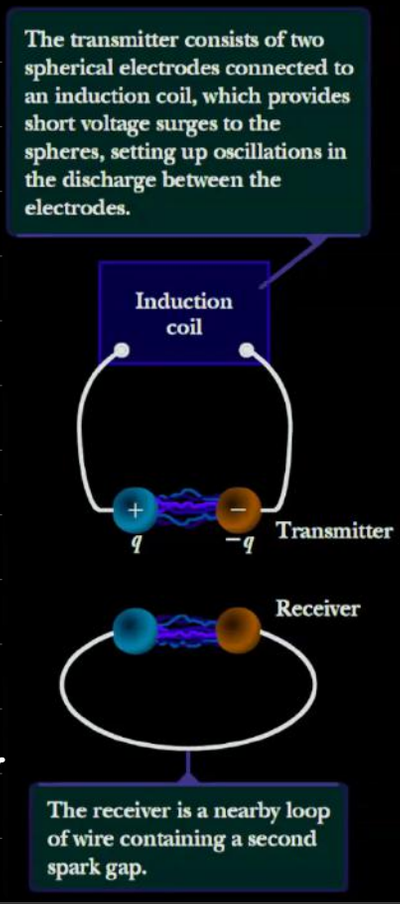


How changing $\vec{E}$ and $\vec{B}$ fields propagate through space.

Hertz Experiment And Discovery Of EM Waves:



The apparatus used by Hertz in 1887 to generate and detect electromagnetic waves.

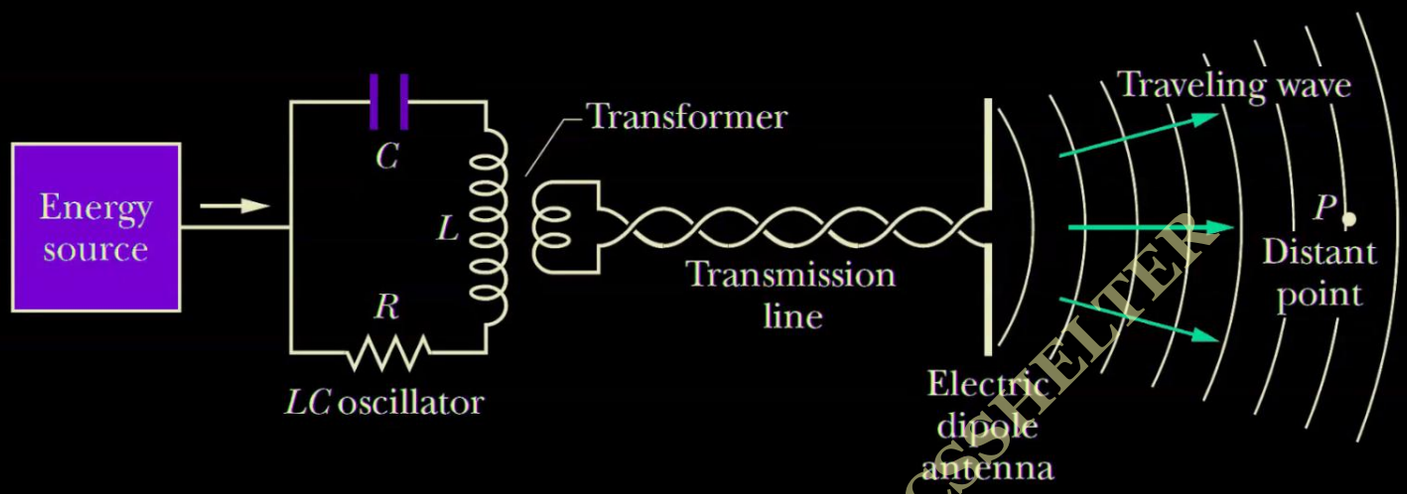


Using Maxwell's Equations, one can predict existence of electromagnetic waves which move at speed of light i.e. c. Hertz performed several experiments which verified Maxwell's prediction. When a high voltage was applied across the transmitting coil, it was seen that a spark jumped across it (see figure above) and almost instantaneously a spark jumped in the receiver coil. Therefore some invisible waves (or energy) must be travelling through space for this kind of action to be observed at some distance. Maxwell found that these waves refracted, reflected and formed stationary waves just like the visible light and could also

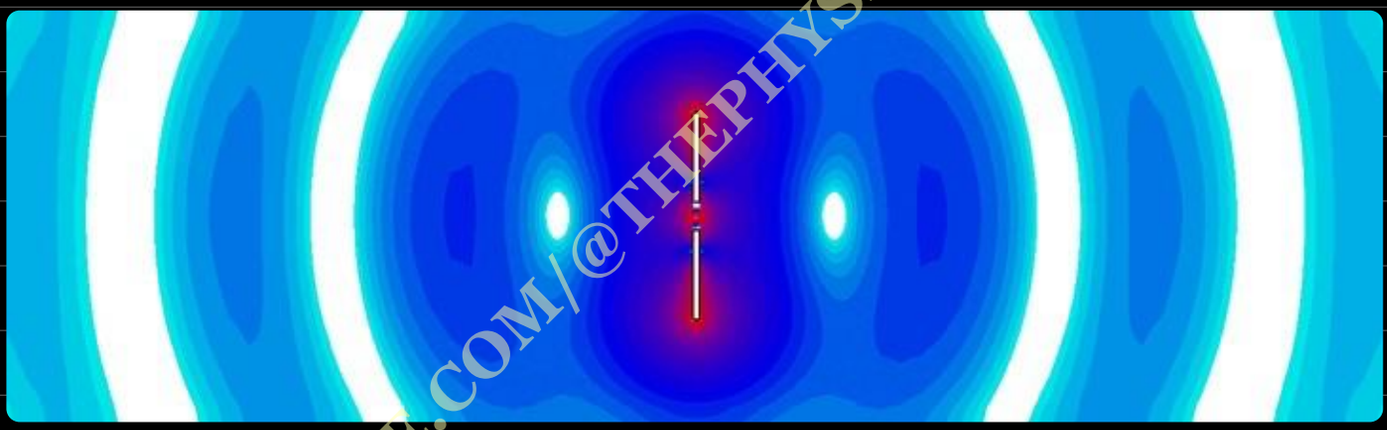
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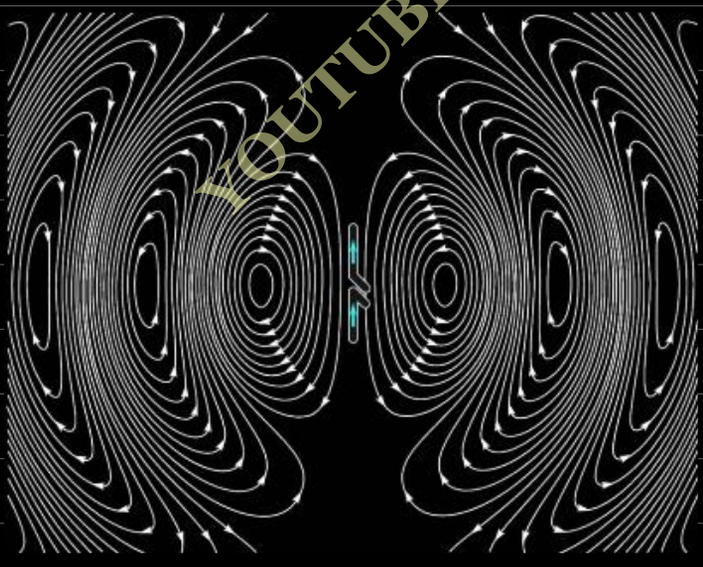
be polarized. These waves were formally called EM waves and visible light happens to be just another form of Electromagnetic waves. (These waves also underwent interference and diffraction just like visible light and also travelled exactly at same speed c ($3\times10^8 ms^{-1}$)).

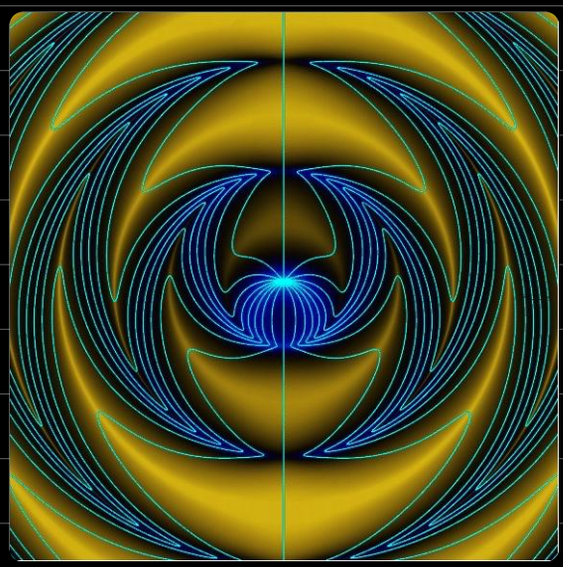
EM WAVE GENERATION FROM DIPOLE ANTENNA:



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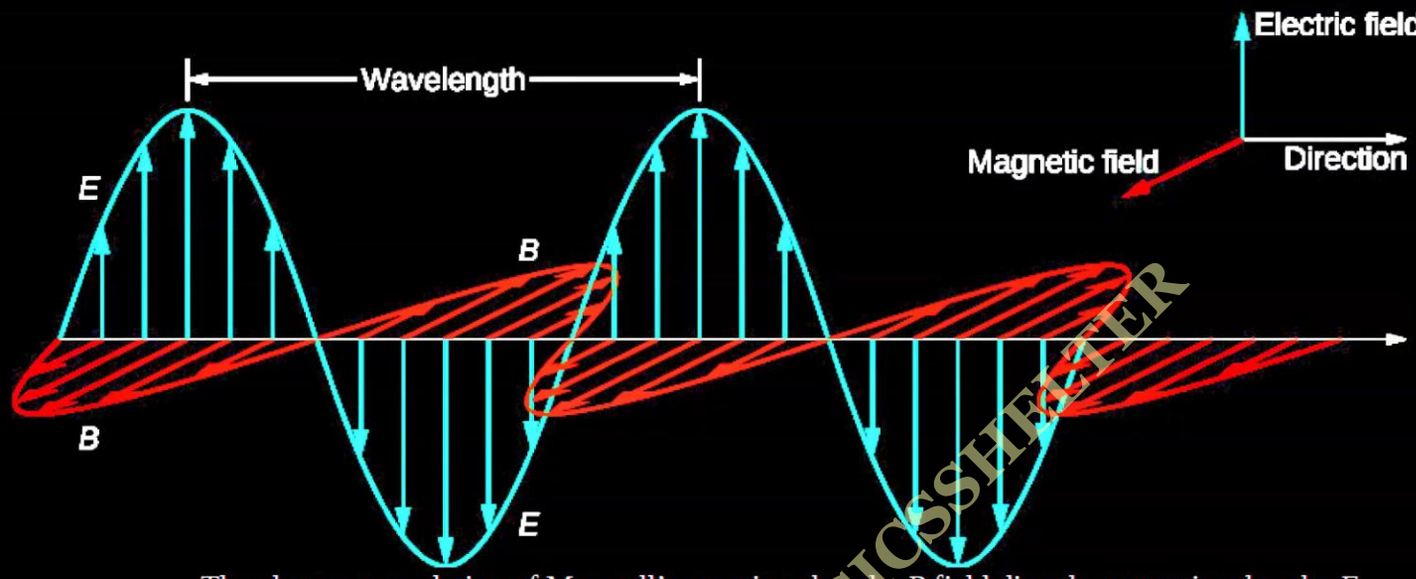




→ A dipole antenna generates EM radiation by creating an electric dipole with periodically oscillating dipole moment. Maximum energy is radiated in direction ⊥ to direction of oscillation.

EM Waves (Qualitative):

At sufficiently large distances from sources, the wavefronts of the wave can be approximated as planar and for small regions we may treat the wave as a plane wave.



The plane wave solution of Maxwell's equations has the B field directly proportional to the E field at each point, with the relative directions shown.

The figure above shows variation of electric and magnetic fields at various points located along a ray in direction of propagation. The followings points are worth noting:

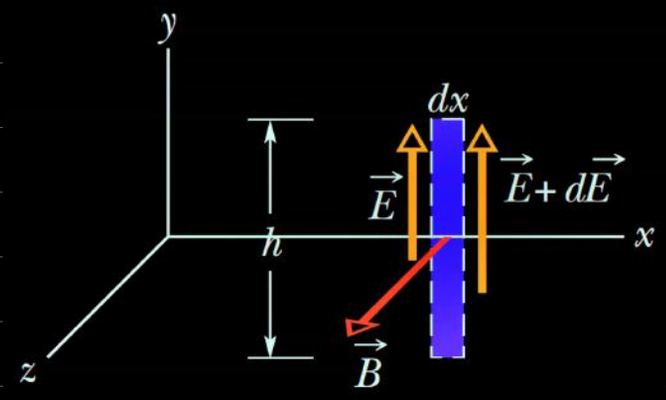
- The electric field and the magnetic field are both perpendicular to the direction of propagation thus the wave is a transverse wave
- Electric field is always perpendicular to the magnetic field.
- The wave travels along the direction $\hat{E} \times \hat{B}$ or $\hat{E} \times \hat{B} = \hat{C}$ (By cyclic exchanges, $\hat{C} \times \hat{E} = \hat{B}$ and $\hat{B} \times \hat{C} = \hat{E}$, just like $\hat{i} \times \hat{j} = \hat{k}$ etc).
- We have considered here a sinusoidal wave, these fields vary with same frequency and wavelength and are in phase with each other (Maxima and minima of both fields occur at same points / times).
- If we assume $E = E_m \sin(kx - \omega t)$ and $B = B_m \sin(kx - \omega t)$ then, $E_m / B_m = E/B = \omega/k = C = 1/\sqrt{\mu_0 \varepsilon_0}$ (This will be proved later on).

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TRAVELING EM WAVE (QUANTITATIVELY):

Consider an EM wave traveling along x axis with electric field oscillations along y direction and magnetic field oscillations along z direction. Now consider the Faraday loop as shown in the adjacent figure. Applying Faraday's law,

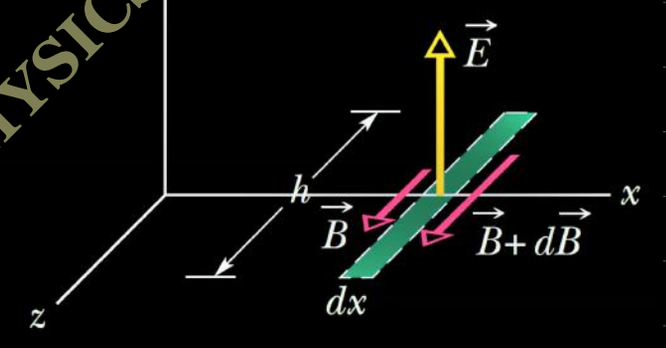
$\int \vec{E}\cdot d\vec{l} = -d\phi_B/dt$ we choose



the loop normal along $\hat{k}$. Thus, we get: $(E+dE)h - Eh = -h\,dx\,\frac{\partial B}{\partial t}$

$$\Rightarrow \boxed{\frac{\partial E}{\partial x} = -\frac{\partial B}{\partial t}} \quad (1)$$

Now consider an Amperian loop with normal along $\hat{j}$ as shown in the figure. Applying Ampere's law for free space:



$$\int \vec{B}\cdot d\vec{l} = \mu_0 \epsilon_0 \frac{d\phi_E}{dt} = \frac{1}{c^2}\frac{d\phi_E}{dt}$$

Thus: $Bh - (B+dB)h = \frac{1}{c^2}\cdot h\,dx\left(\frac{\partial E}{\partial t}\right)$

$$\Rightarrow \boxed{\frac{\partial B}{\partial x} = -\frac{1}{c^2}\frac{\partial E}{\partial t}} \quad (2)$$

Now differentiating (1) wrt x and (2) wrt time we get:

$\frac{\partial^2 E}{\partial x^2} = \frac{-\partial^2 B}{\partial x \partial t}$ (3) and $\frac{\partial^2 B}{\partial x \partial t} = -\frac{1}{c^2}\frac{\partial^2 E}{\partial t^2}$ (4)

From (3) & (4):

$$\boxed{\frac{\partial^2 E}{\partial t^2} = c^2\frac{\partial^2 E}{\partial x^2}}$$

→ This equation looks just like differential equation governing one dimensional waves i.e $\frac{\partial^2 y}{\partial t^2} = v^2\frac{\partial^2 y}{\partial x^2}$, thus speed of our EM wave must be c !!!

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→ But in which frame? We didn't choose any particular frame for applying Maxwell's equations and Maxwell's equations are valid in every inertial frame ∴ speed of EM Waves in free space must be c for every observer. This would lead to complete failure of the Galilean Relativity and marks the beginnings of Einsteinian Relativity.

→ The student should memorize by heart the equations ① and ② for solving for the fields of a wave traveling in one of the standard directions. For a wave traveling along arbitrary direction, The following results will come handy:

$$\vec{\nabla} \times \vec{B} = \frac{1}{c^2}\frac{\partial E}{\partial t} \quad \text{and} \quad \vec{\nabla} \times \vec{E} = -\partial B/\partial t$$

Where $\vec{\nabla}$ represents the operator $\vec{\nabla} = \left[\hat{i}\,\partial/\partial x + \hat{j}\,\partial/\partial y + \hat{k}\,\partial/\partial z\right)$

→ Remember that for a wave traveling along a direction $\vec{k}$, the wave equation is:

$$\vec{E} = E_m \cos(\vec{k}\cdot\vec{r} - \omega t + \phi)\,\hat{E} \quad \text{and}$$
$$\vec{B} = B_m \cos(\vec{k}\cdot\vec{r} - \omega t + \phi)\,\hat{B}$$

EXAMPLE: The electric field component of a monochromatic radiation is given by $\vec{E} = 2E_0 \cos\omega t \cos kz\,\hat{i}$ what is the expression for its magnetic field?

Solution: This is a stationary wave of electric field along z direction.

Now $\vec{\nabla} \times \vec{E} = -\partial\vec{B}/\partial t$

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ 2E_0\cos kz\cos\omega t & 0 & 0 \end{vmatrix} = -\partial\vec{B}/\partial t = -\hat{j}\,[2E_0 k \sin kz \cos\omega t]$$

$$\Rightarrow \vec{B} = \frac{2E_0 k}{\omega}\sin kz \sin\omega t\,\hat{j} \text{ Ans.}$$

EXAMPLE: Electric field of a plane EM Wave is given as

$\vec{E} = \frac{E_0}{\sqrt{2}}(\hat{i}+\hat{j})\cos(kz + \omega t)$, find the magnetic field wave.

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Solution: $\vec{\nabla} \times \vec{E} = -\partial \vec{B}/\partial t$. Thus:

$$\frac{E_0}{\sqrt{2}} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial z & \partial/\partial y \\ \cos(kz+\omega t) & \cos(kz+\omega t) & 0 \end{vmatrix} = -\frac{\partial \vec{B}}{\partial t}$$

$$\Rightarrow \frac{E_0}{\sqrt{2}} \left[\hat{i}\,[k \sin(kz+\omega t)] - \hat{j}\,[k \sin kz+\omega t] \right] = -\partial \vec{B}/\partial t$$

$$\Rightarrow \vec{B} = -\frac{E_0}{\sqrt{2}} [\hat{i} - \hat{j}] \left[-\frac{k}{\omega} \cos(kz+\omega t) \right] = \frac{E_0}{\sqrt{2}c} [\hat{i} - \hat{j}] \cos(kz+\omega t)$$

VELOCITY OF EM WAVES IN DIELECTRIC MEDIA: Just as for free space, the speed of EM wave is given as $c = 1/\sqrt{\mu_0 \epsilon_0}$ similarly for a material medium, speed of the EM wave is given as:

$v = 1/\sqrt{\mu \epsilon}$ where $\mu = \mu_0 \mu_r$ ($\mu_r \rightarrow$ relative permeability) and $\epsilon = \epsilon_0 \epsilon_r$ ($\epsilon_r \rightarrow$ Dielectric constant). Now for non-ferromagnetic materials, $\mu_r \approx 1$ thus $v = \frac{1}{\sqrt{\mu_0 \epsilon_0 \epsilon_r}} = \frac{c}{\sqrt{\epsilon_r}} = \frac{c}{\sqrt{k}}$

thus $\boxed{v = \frac{c}{\sqrt{k}}}$ where k is the dielectric constant.

→ For JEE purposes we do not consider the complex nature of dielectric constant, however actually dielectric constant can be a complex number because field induced in a dielectric may not be exactly in phase with applied field and the phase difference may depend on frequency of the applied field.

→ For non-ferromagnetic materials, the displacement current is given by $i_d = \epsilon\, d\Phi_E/dt$ instead of $\epsilon_0\, d\Phi_E/dt$. Thus Ampere Maxwell law gets modified to:

$$\int \vec{B} \cdot d\vec{\ell} = \mu \left[i + \epsilon_0 K \frac{d\Phi_E}{dt} \right] \qquad (\mu \approx \mu_0).$$

→ Upon refraction, frequency remains same, but wavelength changes.

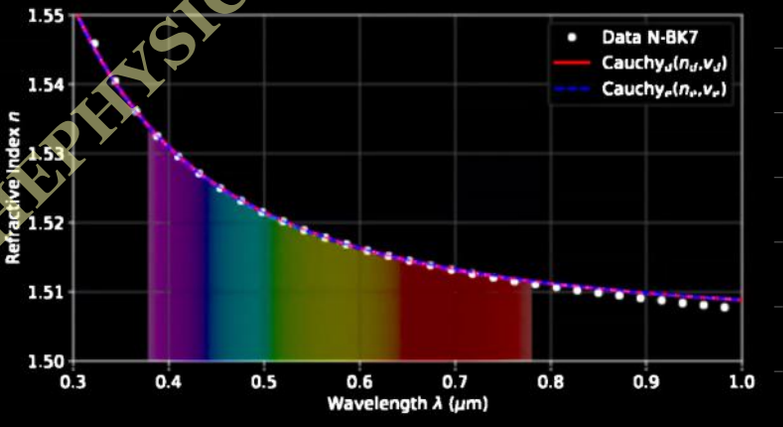
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→ Refractive index of any material is given by $n = c/v$ thus for non-ferromagnetic materials, refractive index is simply $n = \sqrt{\kappa}$ where κ is dielectric constant.

→ κ is a frequency dependent constant. For instance static dielectric constant for water is nearly 80, thus refractive index of water would have been around 9 if dynamic κ were also 80. However for visible frequencies, the refractive index is only about 4/3.

→ The phenomenon of dispersion can be observed in the transparent dielectrics because of frequency dependent dielectric constant. For visible light, the refractive index is given by $n(\lambda) = A + B/\lambda^2 + C/\lambda^4 + \ldots$ which is approximated as $\boxed{n(\lambda) = A + \frac{B}{\lambda^2}}$

→ A typical graph of variation of refractive index of glass with wavelength near visible wavelengths is shown in adjacent figure.
(Though over a wider range of frequencies the graph is not strictly monotonic in general).



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Energy Density Of EM Waves In Space:

Recall that energy density associated with Electric field is $\frac{1}{2}\epsilon_0 E^2$ while that with magnetic field is $B^2/2\mu_0$. Thus energy density associated with EM waves in general is:

$$u = u_E + u_B = \frac{1}{2}\epsilon_0 E^2 + \frac{B^2}{2\mu_0}$$

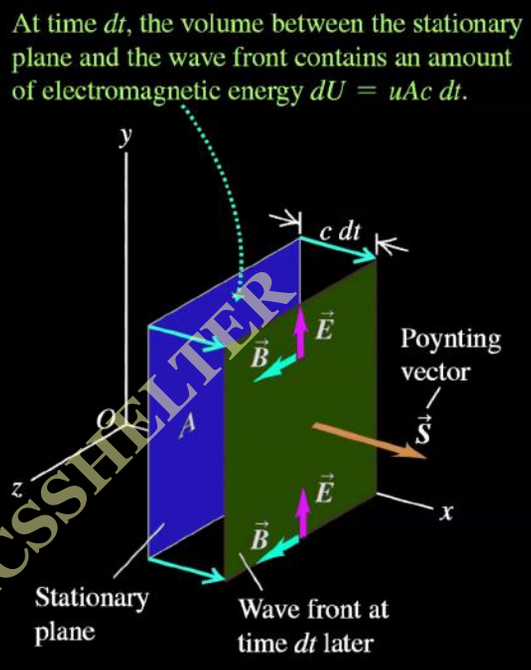
Now for EM waves, $E/B = c$ thus $\frac{1}{2}\epsilon_0 E^2 = \frac{1}{2}\epsilon_0 (Bc)^2 = \frac{1}{2}\epsilon_0 B^2 \cdot \frac{1}{\mu_0\epsilon_0}$

thus $u_E = u_B = \frac{B^2}{2\mu_0} = \frac{1}{2}\epsilon_0 E^2$, thus:

$$\boxed{u = \epsilon_0 E^2 = \frac{B^2}{\mu_0} = \frac{1}{2}\epsilon_0 E^2 + \frac{B^2}{2\mu_0} = \frac{EB}{c\mu_0} = \frac{|\vec{E}\times\vec{B}|}{c\mu_0}}$$

and $\boxed{U_E = U_B = U/2}$

Electromagnetic Energy Flow And Poynting Vector:



Consider a plane wave travelling along x axis with speed c having energy density u. The distance travelled by the wave in a time dt is cdt, thus energy contained in a volume Acdt must have crossed a plane of Area A (see figure). Now if u is the energy density then crossed Energy:

$dU = uAcdt$, Now intensity is defined as energy crossing per unit time per unit area, thus: $I = \frac{dU}{Adt}$ or

$\boxed{I = uc}$

Poynting Vector: A vector whose direction is same as the direction of wave propagation and magnitude equal to Intensity of the wave is called Poynting Vector [Or Umov Vector for Russian physicists]. Thus we have the Poynting vector as:

$$\vec{I} = uc\hat{c} = u\vec{c} \quad \text{or} \quad \boxed{\vec{I} = u\vec{c}}$$

Now we can substitute any expression for u from the previous section to express Poynting vector in terms of desirable fields. One of the common forms uses $u = |\vec{E}\times\vec{B}|/c\mu_0$ thus.

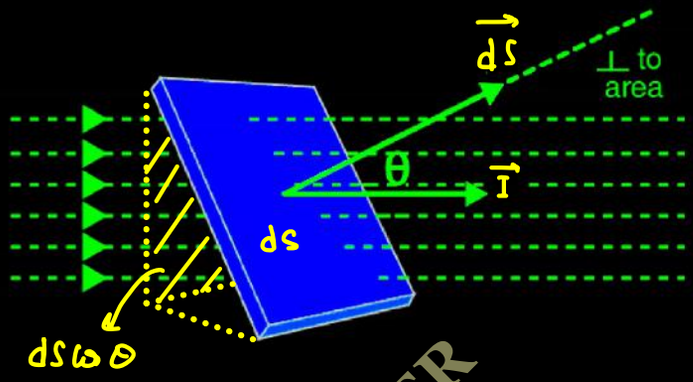
$$\vec{I} = \frac{|\vec{E}\times\vec{B}|}{c\mu_0}\vec{c} = \frac{|\vec{E}\times\vec{B}|}{\mu_0}\hat{c} = \frac{\vec{E}\times\vec{B}}{\mu_0} \quad (\because \hat{c} = \hat{E}\times\hat{B}).$$

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Thus we have the Poynting vector: $\vec{I} = (\vec{E} \times \vec{B})/\mu_0$

FLUX OF POYNTING VECTOR:

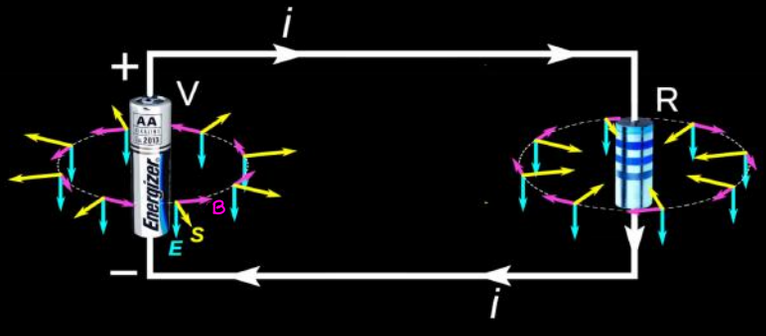
Consider a patch of elemental area $\vec{dS}$, Just as we imagined field lines for Electric fields, we can also imagine field lines for poynting vector and as we understood for electrical flux, here also the flux only depends on projection of patch $\perp$ to the poynting vector. Thus flux of poynting vector for this patch is simply $d\Phi_I = \vec{I} \cdot \vec{dS}$ which simply represents energy crossing the patch per unit time. Thus total electromagnetic Power flow in to or out of any volume is simply:



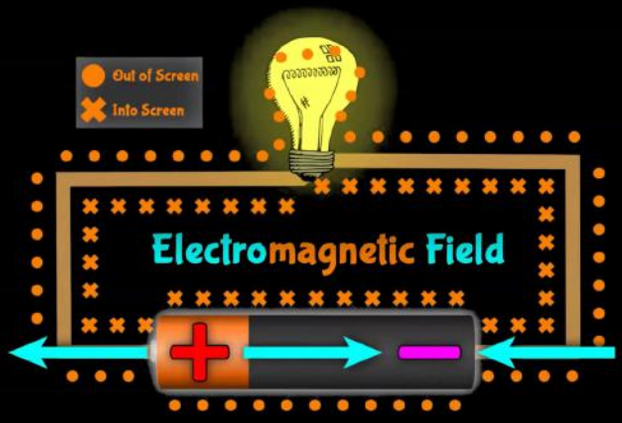
$$P = \oint \vec{I} \cdot \vec{dS}$$

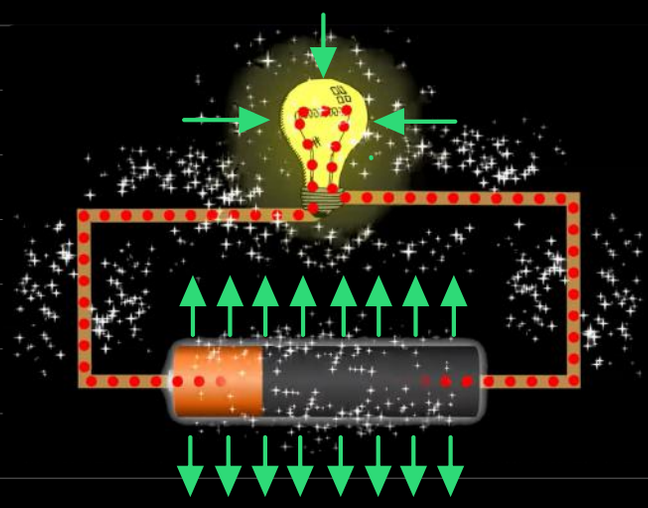
ENERGY FLOW IN ELECTRICAL CIRCUITS:

We might be inclined to think that energy in electrical circuits flows along the wire, however it turns out that while wires are instrumental in energy flow, actual energy flow happens through space. The adjacent figure shows the Electromagnetic field near a battery and a resistor. The energy flow occurs along the direction $\vec{E} \times \vec{B}$, thus from battery the energy flows in to space and from space the energy is flowing in to the resistor which is ultimately dissipated as heat. For a more visual explanation of the phenomena, the students may refer to a video on Poynting Vector by "The Science Asylum" youtube channel. (Following two images are excerpted from the video for understanding the phenomenon).



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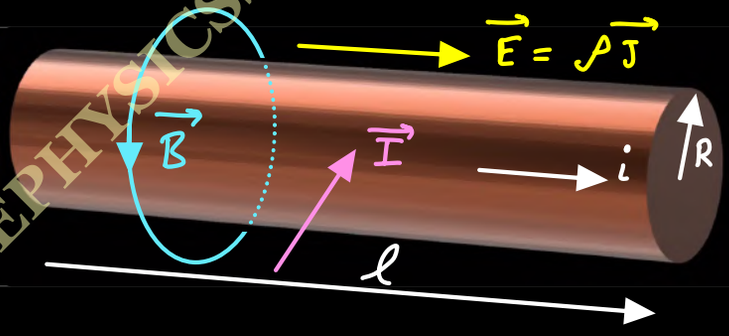


EXAMPLE: IRODOV 4.204:

A current I flows along a straight conductor with round cross-section. Find the flux of the Poynting vector across the lateral surface of the conductor's segment with resistance R.

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Solution: As can be seen in adjacent figure, The axial electric field just outside the conductor is equal to that inside the conductor. Thus axial electric field is simply $\vec{E} = \rho \vec{J}$ where $\vec{J}$ is the current density. Also magnetic field on surface is $\mu_0 i/2\pi R$. Thus poynting vector:



$$\vec{I} = \frac{\vec{E} \times \vec{B}}{\mu_0} = \rho J \times \frac{\mu_0 i}{2\pi R} \cdot \frac{1}{\mu_0} (-\hat{r}) = [\rho J i / 2\pi R][-\hat{r}]$$

Thus net flux of poynting vector over curved surface:

$$P = 2\pi R \ell \times I = \rho \frac{J i}{2\pi R} \times 2\pi R \ell = \rho \ell J i = \rho \ell \frac{i}{A} \cdot i$$

$$\Rightarrow P = i^2 R \text{ Ans.}$$

→ This is the familiar result for power dissipation in a resistor. The Electromagnetic Energy is entering the resistor at this rate and converted to thermal energy which is again finally radiated out to the space.

MOMENTUM FLOW IN EM WAVES: Recall from your study of atomic structure (chemistry) that for a photon, $\lambda = h/p$ and $E = h\nu$ also $\lambda \times \nu = c$ thus combining the three equations we can simply write for a photon, $E = pc$. Now poynting vector gives us the energy flow per unit time per unit area, thus momentum flow per unit time per unit area is simply $1/c$ times the poynting vector, i.e:

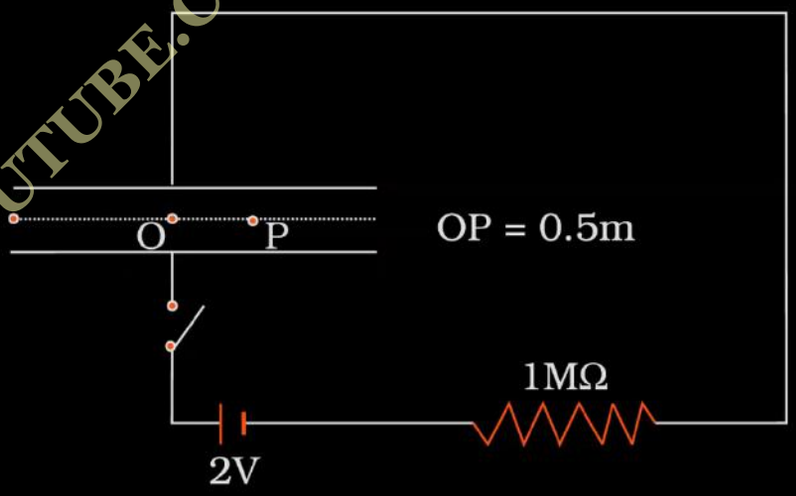
$$\boxed{\vec{u} = \vec{I}/c}$$

Here we are using the symbol $\vec{u}$ for momentum flow per unit time per unit area because it has dimensions of Energy per unit volume and also numerically equal to it. (Recall, $\vec{I} = u\vec{c}$).

→ It also has dimensions of pressure.

EXAMPLE: NCERT: 8.1

Example 8.1 A parallel plate capacitor with circular plates of radius 1 m has a capacitance of 1 nF. At $t = 0$, it is connected for charging in series with a resistor $R = 1\ M\Omega$ across a 2V battery (Fig. 8.3). Calculate the magnetic field at a point P, halfway between the centre and the periphery of the plates, after $t = 10^{-3}$ s. (The charge on the capacitor at time t is $q(t) = CV[1 - \exp(-t/\tau)]$, where the time constant τ is equal to CR.)



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Solution: Consider amperean loop of radius r and let plate radius be a.

Then: $B \times 2\pi r = \mu_0 \epsilon_0 \frac{d}{dt}\left[\frac{\sigma}{\epsilon_0} \times \pi r^2\right]$

$\Rightarrow \quad B = \frac{\mu_0 r}{2} \frac{d\sigma}{dt} = \frac{\mu_0 r}{2} \cdot \frac{1}{\pi a^2} \left(\frac{dq}{dt}\right)$ ans.

For our case, $r = a/2$. Substituting the values, we will get $B = 0.74 \times 10^{-13}$ T.

EXAMPLE: NCERT:

Example 8.3 The magnetic field in a plane electromagnetic wave is given by $B_y = (2 \times 10^{-7})$ T sin $(0.5\times10^{3}x+1.5\times10^{11}t)$.

(a) What is the wavelength and frequency of the wave?

(b) Write an expression for the electric field.

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Solution: (a) $\omega = 1.5 \times 10^{11}$, $k = 0.5 \times 10^{3} = 2\pi/\lambda \Rightarrow \lambda = 2\pi/0.5 \times 10^{3}$

$\Rightarrow \lambda = 1.26$ cm ans.

(b) Recall that:

$$\frac{\partial B}{\partial x} = \frac{-1}{c^2} \frac{\partial E}{\partial t} \quad \text{and} \quad \frac{\partial E}{\partial x} = -\frac{\partial B}{\partial t}$$

Similar expressions also hold in other directions. Now our wave is travelling along $-\hat{i}$ i.e $\hat{c} = -\hat{i}$ and $\hat{B} = \hat{j}$, Now, $\hat{E} \times \hat{B} = \hat{c} \Rightarrow \hat{B} \times \hat{c} = \hat{E} \Rightarrow \hat{E} = \hat{k}$. Also $E/B = c$

$\Rightarrow E_m = cB_m = 3 \times 10^{8} \times 2 \times 10^{-7} = 60$, Also we know that electric field and magnetic field of a travelling EM wave are in same phase, so even without solving the PDE we can directly write $\vec{E} = E_m \sin(kx + \omega t)\hat{k} = 60 \sin(0.5 \times 10^{3} x + 1.5 \times 10^{11} t)\hat{k}$ ans.

AVERAGE INTENSITY AND ENERGY DENSITY OF SINUSOIDAL EM WAVES:

Recall the expressions for energy density: $U_E = U_B = U/2$.

Now let $E = E_m (\sin kx - \omega t)$ then,

$U_E = \frac{1}{2} \epsilon_0 E^2 = \frac{1}{2} \epsilon_0 E_m^2 \sin^2(kx - \omega t)$, thus average electric

energy density $\langle U_E \rangle = \frac{1}{2} \epsilon_0 \langle E^2 \rangle$ i.e. $\frac{1}{2} \epsilon_0 E_{rms}^2 = \frac{1}{2} \epsilon_0 \left(\frac{E_m}{\sqrt{2}}\right)^2$

thus $\langle U_E \rangle = \frac{1}{4} \epsilon_0 E_m^2$

Similarly $\langle U_B \rangle = \frac{1}{4} \frac{B_m^2}{\mu_0}$, Thus:

$$\langle u \rangle = \frac{1}{2} \epsilon_0 E_m^2 = \frac{B_m^2}{2\mu_0} = \frac{1}{2} \frac{E_m B_m}{c \mu_0} = |\vec{u}| \qquad (\because \frac{E_m}{B_m} = c)$$

Now $I = uc \Rightarrow$ $$I = \frac{1}{2} \frac{E_m B_m}{\mu_0} = \frac{B_m^2 c}{2\mu_0} = \frac{1}{2} \epsilon_0 E_m^2 c$$

→ Recall that energy density is also equal to momentum flow per unit area per unit time.

→ Also recall from your study of A.C. that there $E_{rms} = E_m/\sqrt{2}$ and $B_{rms} = B_m/\sqrt{2}$.

→ When nothing is specified then field of an EM wave usually means RMS field.

EXAMPLE: NCERT:

Example 8.5 Calculate the electric and magnetic fields produced by the radiation coming from a 100 W bulb at a distance of 3 m. Assume that the efficiency of the bulb is 2.5% and it is a point source.

Solution: $I = \frac{P}{4\pi r^2} \times \eta = \frac{1}{2} \epsilon_0 E_m^2 \times c$ where η is efficiency

Now $E_m = \sqrt{2}\, E_{rms}$, thus $\frac{\eta P}{4\pi r^2} = c \epsilon_0 E_{rms}^2 = \frac{c B_{rms}^2}{\mu_0}$

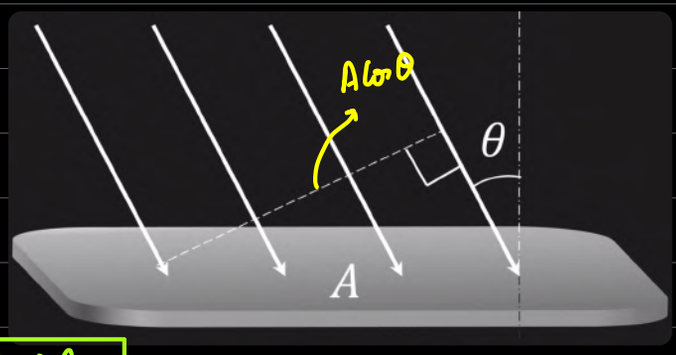
$$\Rightarrow E_{rms} = \sqrt{\frac{\eta P}{4\pi \epsilon_0 c r^2}} = \sqrt{\frac{2.5}{100} \times \frac{100}{3\times10^8 \times 9} \times 9\times10^9} = 5/\sqrt{3} \approx 2.9 \text{ V/m.} \quad \text{Ans.}$$

Now $E_{rms}/B_{rms} = c \Rightarrow B_{rms} = E_{rms}/c \approx 9.7 \times 10^{-9}$ T Ans.

Now $E_m = \sqrt{2}\, E_{rms} = 4.07$ V/m and $B_m = \sqrt{2}\, B_{rms} = 1.4 \times 10^{-8}$ T Ans.

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Radiation Pressure And Shear Stress On Perfectly Absorbing Surface:



Consider a laminar patch of Area A on which radiation falls at an angle θ. Now $\vec{u}$ is the momentum crossing per unit time per unit $\perp$ Area.

Thus momentum falling on the patch per unit time, $\vec{F} = \vec{u} A \cos\theta \Rightarrow \boxed{\vec{F} = -\vec{u}\cdot\vec{A}\,\hat{c}}$

since all the momentum is getting absorbed, this must also be the force on the lamina. The normal component of this force per unit area is called pressure and tangential component is called the shear stress. Thus, the radiation pressure:

$$P_n = F\cos\theta/A = u\cos^2\theta = I/c\cos^2\theta$$

$$\boxed{P_n = \frac{I}{c}\cos^2\theta = u\cos^2\theta}$$

Shear stress, $\sigma_s = \frac{F\sin\theta}{A} = u\cos\theta\sin\theta = \frac{I}{c}\cos\theta\sin\theta$

Thus shear stress: $\boxed{\sigma_s = \frac{I}{c}\cos\theta\sin\theta = u\cos\theta\sin\theta}$

→ If instead of photon beam we have a beam of material particles having Kinetic Energy u per unit volume. Then, energy falling in a time dt would be $u A\cos\theta\, v\, dt$. Now for material particles, momentum and KE are related as $KE = pv/2 \Rightarrow p = 2KE/v$. Thus momentum falling in a time dt will be simply 2/v × Energy falling in a time dt, i.e $2uA\cos\theta\, dt$. Now for force pressure and shear stress, rest of the analysis is same as that for photons. Thus for a beam of material particles.

$$P_n = 2u\cos^2\theta \quad \text{and} \quad \sigma_s = 2u\cos\theta\sin\theta$$

Where u is the Kinetic Energy Density.

→ For Normal incidence, Radiation Pressure, $P_n = I/c$.

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Radiation Pressure On A Perfectly Reflecting Surface:

With a perfectly reflecting surface, change in momentum of photons in the direction normal to the patch will be double that of perfectly absorbing surface, while tangential momentum will remain unchanged. Thus there will be no shear stress, while pressure will be just double the pressure for perfectly absorbing case. Thus here:

$$\boxed{P_r = 2u\cos^2\theta = 2I/c\cos^2\theta} \qquad \sigma_s = 0$$

→ For beam of material particles, u should be replaced with $\frac{1}{2}u$ as discussed in the previous section.

→ In case we have partly reflecting surface with reflectivity η then,

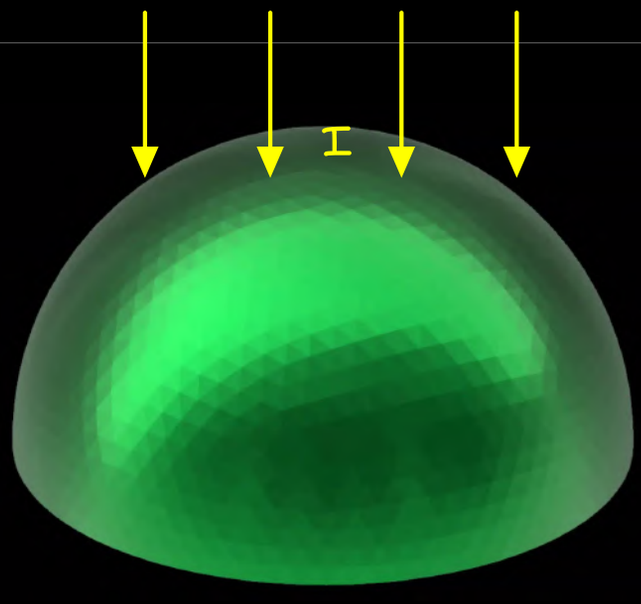
$$P_r = (1-\eta)\frac{I}{c}\cos^2\theta + \eta\frac{2I}{c}\cos^2\theta = (1+\eta)\frac{I}{c}\cos^2\theta$$

$$\sigma_s = (1-\eta)\frac{I}{c}\cos\theta\sin\theta.$$

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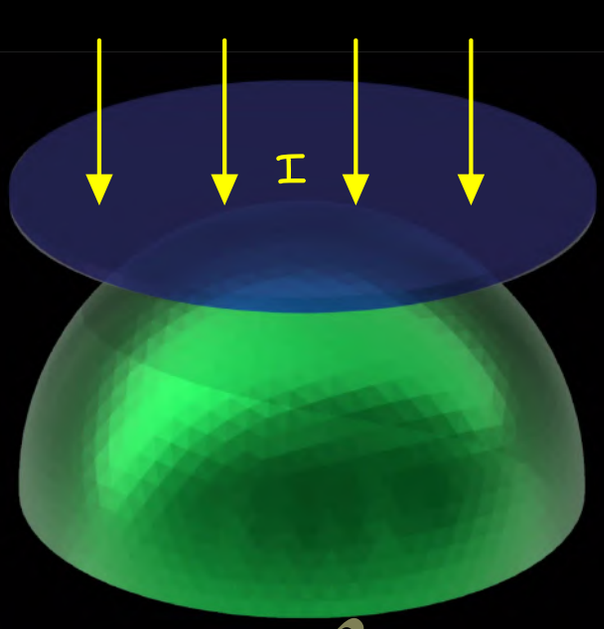
Example:

A radiation beam of intensity I and sufficiently wide aperture falls on a perfectly absorbing hemispherical surface of radius R. Find the net radiation force on the hemisphere.



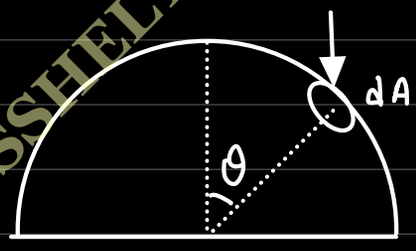
Solution: For absorbing surfaces, the net force only depends on momentum absorbed per unit time. So if we could have an alternate surface which would absorb same momentum per unit time then, force experienced by that surface will also be same.

So here we propose a disc shaped surface which intercepts same energy per unit time as the given hemisphere and hence experiences the same force. Thus $F = \frac{I}{c} \times \pi R^2$ Ans.



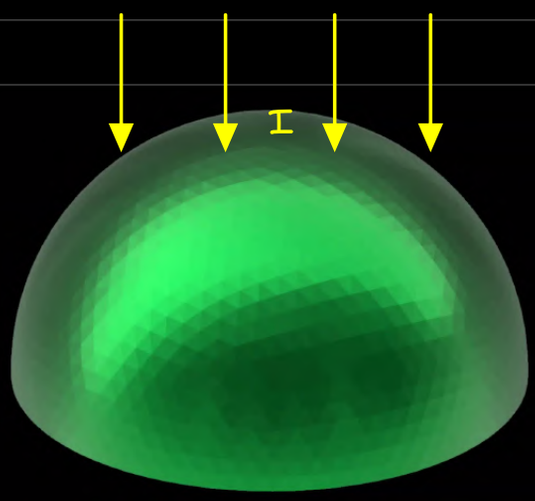
Method 2: Consider a patch of area dA located at an angular position θ as shown in the figure. Now recall that for laminar patches, $F = -\vec{u} \cdot \vec{A}$, thus for this patch, $dF = -\vec{u} \cdot \vec{dA} = u\,dA\cos\theta$, now $dA\cos\theta$ is simply the shadow area of the patch thus, $F = u\int dA\cos\theta = u \times \pi R^2$

Thus $F = \frac{I}{c} \times \pi R^2$ Ans.

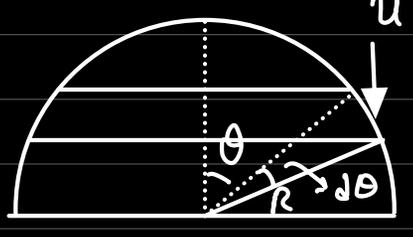


EXAMPLE:

Repeat the previous problem if instead of a perfectly absorbing Hemisphere, you had a perfectly reflecting hemisphere.



Solution: Consider a cap element of the hemisphere as shown in the figure. Pressure on the element is $P_x = \frac{2I}{c}\cos^2\theta$ acting everywhere radially inwards. By symmetry only vertical components of the pressure forces lead to a non zero resultant. Thus elemental vertical force, $dF = P_x \times dA \times \cos\theta$



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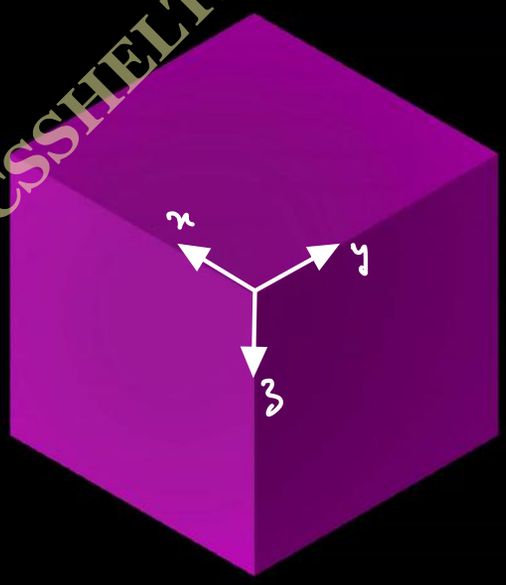
$$\Rightarrow \quad dF = \frac{2I}{c}\cos^2\theta \times 2\pi R\sin\theta \times R\,d\theta \times \cos\theta = \frac{4\pi R^2 I}{c}\cos^3\theta\sin\theta\,d\theta$$

$$\text{Thus } F = \frac{4\pi R^2 I}{c}\int_0^{\pi/2}\cos^3\theta\sin\theta\,d\theta = -\frac{4\pi R^2 I}{c}\left[\frac{\cos^4\theta}{4}\right]_0^{\pi/2} = \frac{\pi R^2 I}{c}\ !!!$$

→ Thus in case of hemisphere, we have same force for perfectly absorbing and perfectly reflecting case (∵ also for partially reflecting case. This should be remembered as a special case).

EXAMPLE:

Light of intensity I is incident along body diagonal of a cube of side ℓ. Find the radiation force on the cube if (a) it is perfectly absorbing (b) It is perfectly reflecting.



Solution: (a) Here we can find force on one of the illuminated faces and multiply it by 3 (due to perfect symmetry). Now For absorbing surface, $F = -\vec{u}\cdot\vec{A} = |-uA\cos\theta|$

Now area normals are along edges while radiation is along body diagonal, and angle between edge and body diagonal is $\cos^{-1}(1/\sqrt{3})$. Thus force on any one surface is $F_1 = u\ell^2 \times \frac{1}{\sqrt{3}}$

Thus net force on cube, $F = 3F_1 = \sqrt{3}u\ell^2 = \sqrt{3}\frac{I}{c}\ell^2$ ans.

(b) Pressure on a reflecting surface is given as $P_n = 2u\cos^2\theta$

Thus pressure force on top illuminated surface, $F_1 = 2u \times \left(\frac{1}{\sqrt{3}}\right)^2 \times \ell^2\hat{k}$

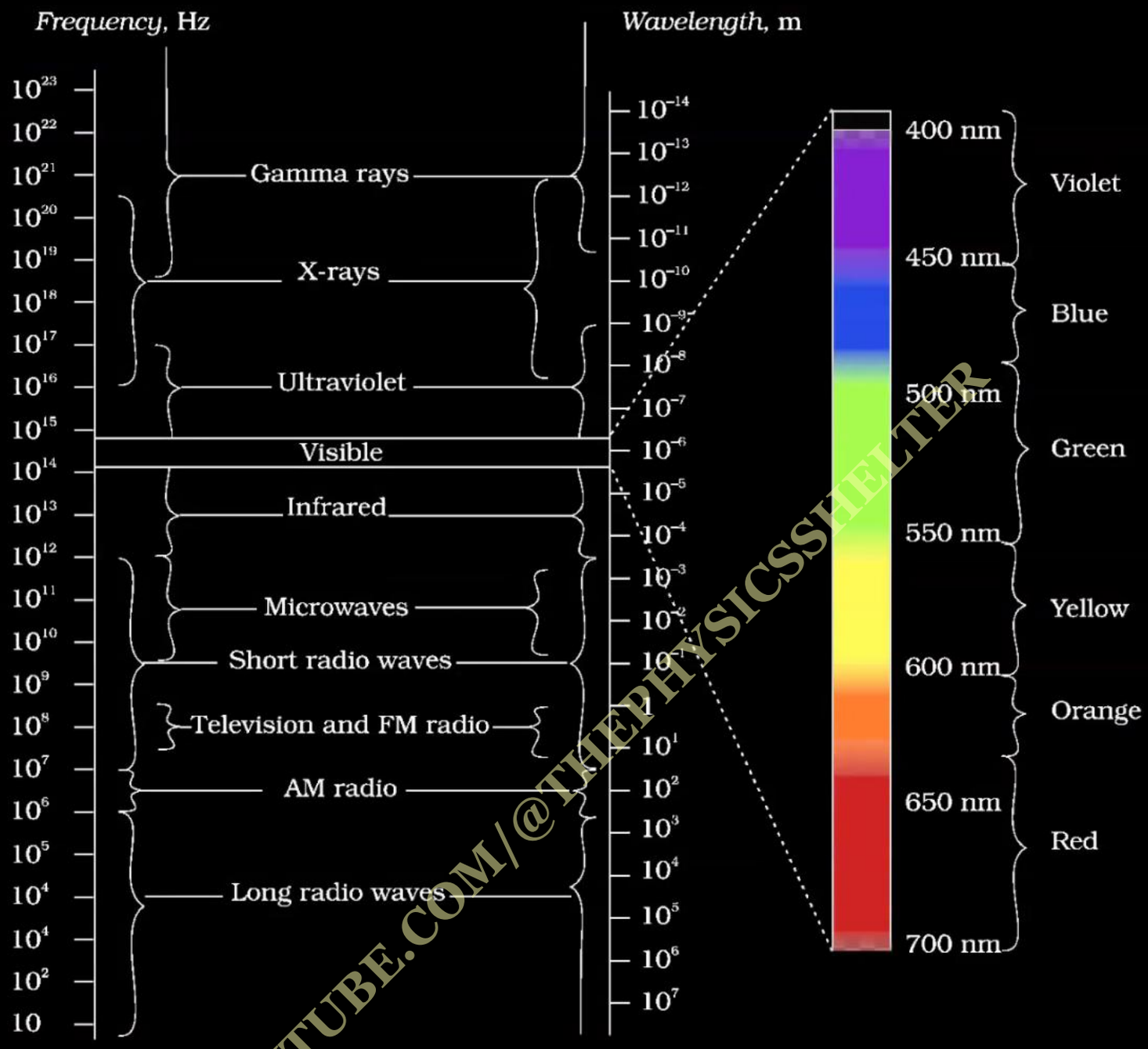
Similarly forces on other two illuminated faces will be in $\hat{i}$ and $\hat{j}$ directions. Thus

$$\vec{F} = \frac{2}{3}u\ell^2\left[\hat{i}+\hat{j}+\hat{k}\right] \text{ ans.}$$

$$F = \frac{2}{\sqrt{3}}u\ell^2 \text{ ans.}$$

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ELECTROMAGNETIC SPECTRUM:



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→ There is no sharp division between one kind of a wave and the next, division is roughly based on how the waves are produced or detected.

Uses Of Em Waves:

Electromagnetic radiation	Uses	
Radio waves	Broadcasting and communications – ~~their longer wavelength means they travel further in the Earth's atmosphere, reflecting off hills and the upper atmosphere~~.	
Microwaves	Cooking food – microwaves are absorbed by water molecules causing them to vibrate (heat up). Satellite transmissions – their wavelength penetrates our atmosphere.	
Infrared	Heater and night vision equipment – all objects, including people, give out infrared rays which can be detected even at night. It's also used for television remote controls.	
Visible light	Human vision, photography and optical fibres – it's the only part of the spectrum we can see.	
Ultraviolet	Fluorescent lamps – they have chemicals inside them which absorb ultraviolet rays and convert the energy to visible light.	
X-rays	Medical equipment – they enable us to see the internal structure of objects and materials by passing through some substances (eg body tissue) but being absorbed by others (eg bone).	
Gamma rays	Sterilising food and medical equipment – they are highly penetrative and can kill.	

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Production And Detection Of Em Waves:

Table 8.1 Different types of electromagnetic waves

Type	Wavelength range	Production	Detection
Radio	> 0.1 m	Rapid acceleration and decelerations of electrons in aerials	Receiver's aerials
Microwave	0.1m to 1 mm	Klystron valve or magnetron valve	Point contact diodes
Infra-red	1mm to 700 nm	Vibration of atoms and molecules	Thermopiles Bolometer, Infrared photographic film
Light	700 nm to 400 nm	Electrons in atoms emit light when they move from one energy level to a lower energy level	The eye Photocells Photographic film
Ultraviolet	400 nm to 1nm	Inner shell electrons in atoms moving from one energy level to a lower level	Photocells Photographic film
X-rays	1nm to 10^{-3} nm	X-ray tubes or inner shell electrons	Photographic film Geiger tubes Ionisation chamber
Gamma rays	$<10^{-3}$ nm	Radioactive decay of the nucleus	-do-

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Em Waves In Dielectric Media: In dielectric media, we substitute ϵ for ϵ_0 and μ for μ_0. Accordingly wave speed becomes $\frac{1}{\sqrt{\mu\epsilon}}$ $= c/\sqrt{k}$ for non-ferromagnetic materials. Also $E/B = v = c/n$ where n is the refractive index. Just as in free space, magnetic and electric energies are equally divided in dielectric media also.

Example: An electromagnetic wave enters a medium of refractive index n without any loss (a) What is the ratio of electric fields before and after the entry. (b) What is the ratio of magnetic fields before and after the entry.

Solution: Since there is no loss, energy incident per unit time must be energy transmitted per unit time. Just as for free space, $I = uc$ similarly in dielectrics $I = uv$ where v

is the velocity in the dielectric media. Thus:

$$\frac{1}{2}\epsilon_0 E_0^2 c = \frac{1}{2}\epsilon E_d^2 v \qquad (d: \text{dielectric}, 0: \text{space}).$$

$$\Rightarrow \frac{E_0}{E_d} = \sqrt{\frac{k}{n}}, \quad \text{now } n = \sqrt{k} \Rightarrow k = n^2 \text{ thus:}$$

$$\frac{E_0}{E_d} = \sqrt{n} \quad \text{Ans.}$$

Similarly,

$$\frac{B_0^2}{2\mu_0} c = \frac{B^2}{2\mu} v \quad \text{but } \mu \approx \mu_0 \Rightarrow \frac{B_0}{B} = \sqrt{\frac{v}{c}} = \frac{1}{\sqrt{n}} \text{ Ans.}$$

EXAMPLE: A light wave is incident normally on a glass slab of refractive index 1.5. If 4% of light gets reflected and amplitude of electric field of incident wave is 30 V/m what is the amplitude of electric field in dielectric medium?

Solution: $(1-\eta)\frac{1}{2}\epsilon_0 E_0^2 \times c = \frac{1}{2}\epsilon E^2 \times v$ (η: Reflected fraction)

Thus: $E = E_0\sqrt{\frac{c}{v}\cdot\frac{1}{k}(1-\eta)} = E_0\sqrt{\frac{1-\eta}{n}} = 30\sqrt{\frac{0.96}{1.5}}$

$$\Rightarrow E = 24 \text{ V/m} \quad \text{Ans.}$$

DISPLACEMENT CURRENT DENSITY IN DIELECTRIC MEDIA:

Recall that in free space, displacement current is given as $\epsilon_0 \frac{d\phi_E}{dt}$ where ϕ_E is electric flux. Now if area of the loop perpendicular to field is A then $\phi_E = EA$ where E is local electric field. Thus $\frac{d\phi_E}{dt} = A\frac{dE}{dt}$, Now displacement current density is simply displacement current per unit area ⊥ to Electric field thus $j_d = \frac{1}{A}\epsilon_0\frac{d\phi_E}{dt}$

i.e $\vec{j}_d = \epsilon_0\frac{d\vec{E}}{dt}$, similarly in dielectric media, $\vec{j}_d = \epsilon\frac{d\vec{E}}{dt}$

where $\epsilon = \epsilon_0\epsilon_r$ (ϵ_r is dielectric constant).

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EXAMPLE: IRODOV 4.191: A plane electromagnetic wave of frequency $\nu = 10\ MHz$ propagates in a poorly conducting medium with conductivity $\sigma = 10\ mS/m$ and dielectric constant $\epsilon_r = 9$. Find the ratio of Amplitudes of conduction and displacement current densities.

Solution: Conduction current density $\vec{j}_c = \vec{E}/\rho = \sigma\vec{E}$

Now displacement current density $\vec{j}_d = \epsilon \frac{d\vec{E}}{dt}$

Now let $\vec{E} = \vec{E}_0 \sin(\omega t - kx)$ then $\frac{d\vec{E}}{dt} = \vec{E}_0 \omega \cos(\omega t - kx)$

Thus Amplitude of j_c $j_{cm} = \sigma E_0$ and Amplitude of displacement current $j_{dm} = \epsilon E_0 \omega$, thus required ratio:

$$j_{cm}/j_{dm} = \sigma/\epsilon\omega = \sigma/(\epsilon_0 \times \epsilon_r \times 2\pi\nu) \text{ Ans.}$$

On putting the values, we get this as 2. (The two are $\pi/2$ out of phase)

EXAMPLE: IRODOV 4.192: A plane EM wave $\vec{E} = \vec{E}_m \cos(\omega t - \vec{k}\cdot\vec{r})$ propagates in free space. Find the corresponding $\vec{B}$ wave and hence find $\vec{B}(t)$ at $\vec{r} = 0$.

Solution: Magnetic field has same functional form as Electric field wave and a direction given by $\hat{B} = \hat{c} \times \hat{E}$. Also $E/B = c$

$$\vec{B} = \frac{E_m}{c}\cos(\omega t - \vec{k}\cdot\vec{r})\,(\hat{c} \times \hat{E}_m) = \frac{1}{c}\hat{k} \times \left[\vec{E}_m \cos(\omega t - \vec{k}\cdot\vec{r})\right]$$

$$= \frac{1}{c}\hat{k} \times \vec{E}, \quad \text{Now @ } \vec{r} = 0, \quad \vec{B}(t) = \frac{1}{c}\hat{k} \times \vec{E}_m \cos(\omega t) \text{ Ans.}$$

→ Thus for travelling EM waves expressed in this form we can quickly write

$$\boxed{\vec{B} = \frac{\vec{k} \times \vec{E}}{kc}} \quad \text{Similarly} \quad \boxed{\vec{E} = \frac{\vec{B} \times \vec{k}}{k}c}$$

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Example: Irodov: 4.196: Find mean Poynting vector $\langle \vec{I} \rangle$ of a plane wave $\vec{E} = \vec{E}_m \cos(\omega t - \vec{k}\cdot\vec{r})$ traveling in space.

Solution: $\vec{I} = \vec{u}\, c$, Now $\langle u \rangle = \frac{1}{2}\epsilon_0 E_m^2$, thus

$$\langle \vec{I} \rangle = \frac{1}{2}\epsilon_0 E_m^2 c \frac{\vec{k}}{k} \text{ Ans.} \qquad (\text{also } c = \omega/k)$$

Think: If an EM Wave is given as $\vec{E} = E_m \sin(\omega t - (3x+4y+5z))$ what is average power crossing a circle $x^2+y^2 = R^2$? [in x-y plane]

Solution:

$$\langle \vec{I} \rangle = \frac{1}{2}\epsilon_0 E_m^2 c \hat{k} = \frac{1}{2}\epsilon_0 E_m^2 c \frac{(3\hat{i}+4\hat{j}+5\hat{k})}{\sqrt{50}}$$

Now power $P_{avg} = \langle \vec{I} \rangle \cdot \vec{A}$, $\vec{A} = \pi R^2 \hat{k} \Rightarrow P = \frac{5\pi R^2}{2\sqrt{50}}\epsilon_0 E_m^2 c$ Ans.

Example: Irodov: 4.198: A ball of radius $R = 50\,cm$ is located in a non magnetic medium with dielectric constant $\epsilon_r = 4$. The amplitude of Electric field in the medium is $E_m = 200\,V/m$. What amount of energy reaches the ball in a time interval $t = 1\,min$.

Solution:

$$|\langle \vec{I} \rangle| = \frac{1}{2}\epsilon E_m^2 v, \quad v = c/\sqrt{\epsilon_r}, \quad \text{Now energy reaching}$$

the ball, $U = I\pi R^2 t = \frac{1}{2}\epsilon_0 \epsilon_r E_m^2 \frac{c}{\sqrt{\epsilon_r}} \pi R^2 t$ Ans.

Now putting the values, $U = 5\,kJ$ (Now $v = c/\sqrt{\mu_r \epsilon_r}$ but for non-magnetic materials, $\mu_r \approx 1$).

Example: Irodov 4.205:

Non-relativistic protons accelerated by a potential difference U form a round beam with current I. Find the magnitude and direction of the Poynting vector outside the beam at a distance r from its axis.

Solution: Recall that $\vec{B} = \frac{\vec{v}\times\vec{E}}{c^2}$ due to a moving charged particle.

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on a translating charged body. Now $\vec{I} = (\vec{E} \times \vec{B})/\mu_0$

$$= \frac{1}{\mu_0 c^2}\left[\vec{E} \times (\vec{v} \times \vec{E})\right] = \frac{1}{\mu_0 c^2}\left[\vec{v}E^2 - \vec{E}(\vec{v}\cdot\vec{E})\right]$$

Now velocity of particles is along the proton beam while $\vec{E} \perp$ to $\vec{v}$. Thus:

$$\vec{I} = \frac{E^2}{\mu_0 c^2}\vec{v}, \text{ Now since } \vec{v} \perp \vec{B} \perp \vec{E} \text{ thus}$$

$\vec{B} = (\vec{v} \times \vec{E})/c^2$ gives $E = \frac{Bc^2}{v}$, thus $\vec{I} = \frac{B^2 c^2}{\mu_0}\vec{v}$

Now $eU = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{\frac{2eU}{m}}$, $B = \frac{\mu_0 I}{2\pi r}$ thus,

$$\vec{I} = \left(\frac{\mu_0 I}{2\pi r}\right)^2 \frac{c^2}{\mu_0}\sqrt{\frac{m}{2eU}}\hat{v} = \frac{I^2}{4\pi^2 r^2 \epsilon_0}\sqrt{\frac{m}{2eU}}\hat{v} \text{ Ans. } \left(\because c^2 = \frac{1}{\mu_0\epsilon_0}\right)$$

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